Treating the Tremors We Built into Robots

A checked motion history, a tracking bound, and the cost of smoothing

Mike Helsel
Working research note; AI assistance disclosed below

17 October 2026 Draft for technical review

Abstract

Frequent target changes can produce unwanted motion even when each plan limits jerk. We study a scalar execution layer based on an established four-stage moving average. A Lean proof shows that its rational history check is equivalent to a bound on reference jerk at every future real time. A finite admission rule retains committed actions and preserves executed motion. A separate theorem connects this reference to a tracking bound for a stated uncertain actuator model.

A fixed comparison shows a benefit and a cost. The filter reduces motion in one hold task, but responds more slowly to a real target change. That hold input aligns with a frequency zero of the filter. A harder, predeclared comparison finds hold failures under irregular and random requests. A duration-limited baseline also removes much of the first apparent advantage. A further Lean theorem bounds tracking across rounded native cubic handoffs. It applies to 12 frozen baseline paths and retains the true-goal failures. We distinguish formal proof, numerical evidence, and open implementation questions. This note does not claim a solution to general robot or VLA jitter.

1 The problem is larger than smooth curves

A planner requests a target. An execution layer turns requests into reference motion. A controller makes the machine follow that motion. These are different objects:

$$\underbrace{g(t)}_{\text{task goal}} \qquad \underbrace{q(t)}_{\text{reference position}} \qquad \underbrace{x(t)}_{\text{physical position}} .$$

Their errors satisfy

$$x(t) - g(t) = [x(t) - q(t)] + [q(t) - g(t)]. \tag{1}$$

A small tracking error $x - q$ does not ensure a small task error $x - g$. The controller can accurately follow a reference that moves away from a stationary goal.

Plain-language contract. Keep motion already committed. Check the whole remaining reference before accepting a request. State the delay that the task can tolerate. Then bound the physical departure from that reference.

The artifact proves part of this contract for a finite scalar history. Requests have separate admission and execution times. The executor does not infer whether a small request is noise or a real change in intent. That distinction needs task information.

The title describes an engineering symptom. It does not imply a neurological mechanism or a common cause of all robot vibration. A coupled arm, contact, sampled control, and a learned policy remain outside this model.

2 An exact check of the retained reference

Let $h > 0$ be one moving-average width. Write $z_+ = \max(z, 0)$ and define

$$F(z) = \frac{z_+^4 - 4(z-1)_+^4 + 6(z-2)_+^4 - 4(z-3)_+^4 + (z-4)_+^4}{24}.$$

This is the step response of four cascaded box averages. For a finite event list $E = ((\tau_i, \delta_i))_{i=1}^n$, use

$$q_E(t) = g_0 + \sum_i \delta_i F\left(\frac{t-\tau_i}{h}\right), \qquad j_E(t) = h^{-3} \sum_i \delta_i K\left(\frac{t-\tau_i}{h}\right).$$

Here $K = F'''$ has branches $0, z, 4-3z, 3z-8, 4-z, 0$ on the consecutive intervals with boundaries $0, 1, 2, 3, 4$. Adjacent formulas agree at the shared boundaries. Lean proves the derivative chain through jerk at every point; q_E is C^3.

At admission time c, form

$$S(E, c) = \{c\} \cup \{\tau_i + kh : \ 1 \le i \le n, \ k = 0, \dots, 4, \ \tau_i + kh \ge c\}.$$

Theorem 1 (Finite check and future jerk)**.** *For any finite list, $h > 0$ and $J \ge 0$,*

$$(\forall s \in S(E, c), \ |j_E(s)| \le J) \quad \Longleftrightarrow \quad (\forall t \ge c, \ |j_E(t)| \le J).$$

For rational inputs, the executable Lean Boolean check is equivalent to this property over all real future times.

Proof idea. Between consecutive roots, jerk is affine. Its absolute value is bounded by its endpoint values. After the last root, jerk is zero. The formal proof constructs the intervals from the event list. It covers an admission time between roots, repeated roots, empty histories, old events, and future events. Signed cancellation remains part of the check.

Admission rule. For admission time c and execution time τ, require $0 \le c \le \tau$, nondecreasing admission times, and τ no earlier than the last committed execution. Express the new increment relative to the final committed target. Append it only if the future check passes. Invalid timing, rejection, and a no-change request retain history.

Finite induction proves the reference-jerk bound from the empty history. New events contribute zero through their execution time. Thus later admissions preserve the executed position, velocity, acceleration, and jerk. After the last execution time plus $4h$, the reference reaches the final committed target at rest. This does not assert physical rest or completion of a rejected goal.

Boundary. The wrapper receives one selected packet. Upstream batch selection and duplicate-conflict handling are not proved. Caller timestamps are not a proof of a real clock. The history is finite; bounded storage, pruning, infinite streams, and measured execution deadlines remain open. Exactness certifies the Lean rational check, not the Python scripts, input parser, upstream scheduler, or native solver. For n events, the direct check visits at most $5n + 1$ points and sums n contributions at each. No real-time execution cost is established.

3 What the physical tracking proof says

The scalar plant is

$$\dot{x} = v, \qquad \dot{v} = a, \qquad \dot{a} = r(t)[u + w(t) - a]. \tag{2}$$

Here u is an acceleration command, w is an additive acceleration disturbance, and r is an actuator response rate. Let $\sigma > 0$, with $8\sigma \le r(t) \le 12\sigma$ and $|w(t)| \le W$.

Use direct feedback

$$u = \ddot{q} - 8\sigma^2 e_x - 6\sigma e_v - \tfrac{1}{2} e_a + \frac{5}{48\sigma} j, \tag{3}$$

where $e_x = x - q$, $e_v = v - \dot{q}$, and $e_a = a - \ddot{q}$. No additional command filter is part of this theorem. Define

$$z_1 = e_x, \quad z_2 = e_x + \frac{e_v}{4\sigma}, \quad z_3 = e_x + \frac{e_v}{2\sigma} + \frac{e_a}{16\sigma^2}.$$

Theorem 2 (Conditional tracking bound)**.** *Fix $H \ge 0$, $R \ge 0$, and $\sigma > 0$. Let $q = q_E$ be the admitted reference above and $j = q_E'''$. Suppose the physical state is a classical trajectory of (2)–(3), continuous on $[0, H]$, with the stated rate and disturbance bounds at every time in $[0, H]$. Suppose $|z_i(0)| \le R$, $|j(t)| \le J$ for every $t \in [0, H]$, and*

$$W + \frac{J}{48\sigma} \le 2\sigma^2 R. \tag{4}$$

Then every $t \in [0, H]$ satisfies

$$|e_x(t)| \le R, \qquad |e_v(t)| \le 8\sigma R, \qquad |e_a(t)| \le 64\sigma^2 R.$$

The composition proof derives the reference derivatives and jerk premise from the admitted history. It applies the earlier scaled invariant-cube theorem to the physical dynamics. The physical state is not reset at reference events.

Illustrative values. Use $\sigma = 10$ s^{-1}, $h = .2$ s, $J = 7.5$ m/s^3, $W = .01$ m/s^2 and $R = .2$ mm. The budget is $.025625 \le .04$ m/s^2. The rate interval is $[80, 120]$ s^{-1}. No robot identification supports these illustrative values.

The theorem bounds $x - q$. A task-position guarantee also needs a bound on $q - g$ in (1). It does not prove physical jerk below 7.5 m/s^3, physical stopping, or a requested deadline. It assumes a physical trajectory exists. Existence for arbitrary discontinuous uncertainty is not proved.

Proof evidence. Independent computational review rebuilt the 44-module local dependency closure and replayed nine modules with the separate Lean checker. The audit covered 106 package theorem statements and 554 inherited statements. They are not 660 research discoveries. Only the standard axioms `propext`, `Classical.choice`, and `Quot.sound` appeared. Four false variants were rejected. These checks support the formal statements, not the physical model's validity.

4 The first comparison: motion versus response time

The first gate has 72 cases: three methods, four synthetic request streams, three constant actuator rates, and two constant disturbances. Each has 1 ms and .5 ms output traces. Raw target changes, native Ruckig replanning, and the guarded four-box reference share the physical feedback law and continuous physical state.

Ruckig uses $|\dot{q}| \le .1$ m/s, $|\ddot{q}| \le .5$ m/s^2 and $|j| \le 7.5$ m/s^3. The FIR uses $h = .2$ s. The general admission theorem checks jerk only; it does not certify every velocity or acceleration limit.

Across six plant settings	Raw	Ruckig	Four-box
Hold: peak $\|x\|$ (mm)	.123–.136	.986–1.014	.0334–.0460
Hold: sampled peak $\|\dot{a}\|$ (m/s^3)	136–209	8.07–11.81	.0364–1.20
Step: absolute settling time (s)	.5035–.504	.6895–.690	.9635–.964

Ranges use the saved .5 ms traces and event samples. Times start at simulation time zero. The step commits at .22 s. Settling means no later sampled violation of $|x - g| \le .2$ mm, $|v| \le .001$ m/s and $|a| \le .02$ m/s^2. The step deadline is 1.5 s and its horizon is 2 s; hold uses 1.9 s and 2.5 s. The revised-goal stream uses 1.75 s and 2.25 s; late/stale uses 1.5 s and 2 s. There is no infinite-time claim.

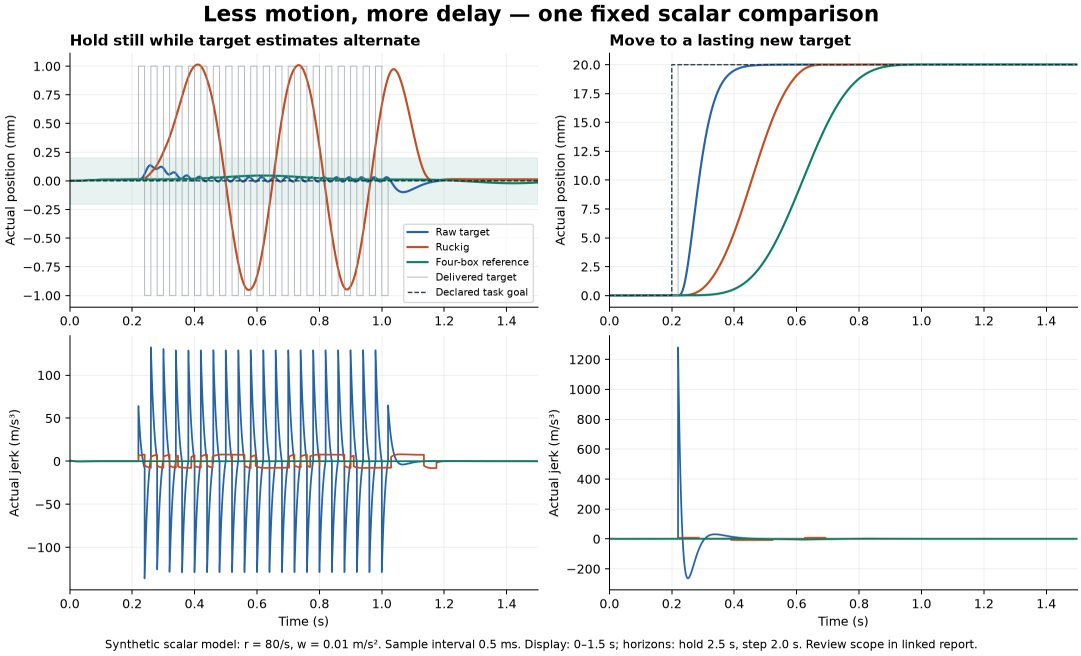


Figure 1. Fixed examples with $r = 80$ s^{-1} and $w = .01$ m/s^2. These curves and their peak metrics are numerical evidence.

An independent 80-digit replay checked all 476,928 trace rows and 3,636 event rows. All 72 endpoint tests passed. Six Ruckig hold cases failed the added .2 mm hold band, leaving 66/72 full-task passes. All 24 FIR and 24 Ruckig cases passed the sampled position band $|x - q| \le .2$ mm. This metric does not test the full transformed cube. At this stage, the Lean history and composition certificate covers FIR only. These Ruckig passes are numerical. Section 7 later proves conditional tracking for the 12 frozen native paths in the repaired comparison, including their exported state roundoff. Raw steps also lie outside that proof contract.

5 The motion in the first hold example

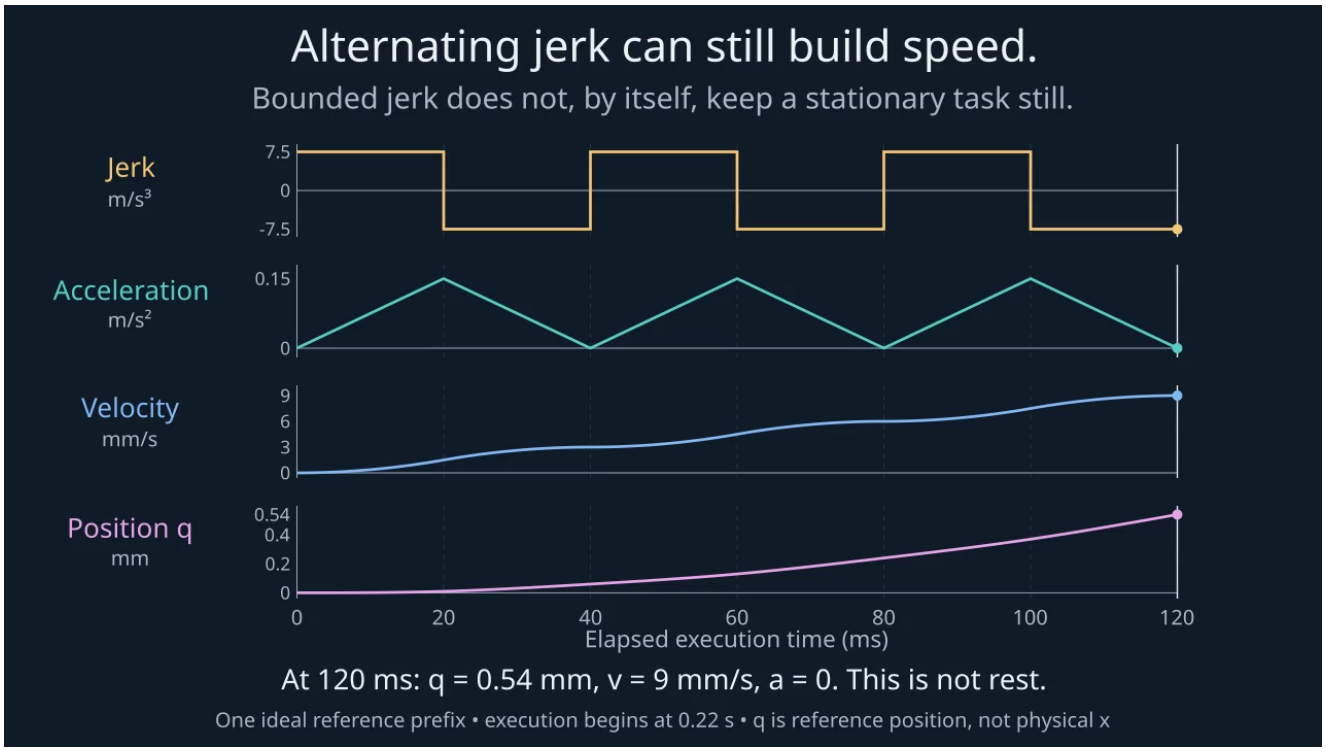


Figure 2. Still from the accompanying Manim animation. It shows an ideal reference prefix, not physical axis motion. Exact segment endpoints match the independently reviewed native states within the recorded numerical tolerance.

The first six executed intervals alternate jerk between $+7.5$ and -7.5 m/s^3. Each lasts 20 ms. Acceleration rises from zero to .15 m/s^2, then returns to zero. It stays nonnegative during this prefix, so velocity accumulates. After 120 ms, the reference has moved .54 mm and still has velocity 9 mm/s.

This explains one trace without a clock error. Later replans change the pattern. It does not establish a defect in Ruckig, which solves a different local planning problem [1]. A bound on each plan's jerk does not by itself keep a stationary task still.

The filter has a favorable input. The target alternates every 20 ms. Its full period is 40 ms. One 200 ms box spans exactly five periods and cancels the sustained alternating signal after its start transient. Four boxes inherit that frequency zero. This example cannot establish general rejection of irregular jitter.

The response is slower. Four boxes have an 800 ms support and a 400 ms mean delay. In a retained negative case, a 20 mm step reaches only 10 mm by a deadline 400 ms after acceptance. A proposed 50 mm change fails the jerk check and is rejected.

The guard does not cause these better curves. Every primary FIR proposal passes. Removing the guard leaves the same reference. The guard rejects an invalid proposal and preserves committed history. Its proof and the filter's numerical performance are separate results.

6 A harder, predeclared comparison

Gate 05 fixes six request streams and four methods before physical runs. It adds Ruckig with minimum duration .8 s at each replan. All 246 such plans return .8 s. This matches the FIR support, but does not ensure equal physical settling time.

The streams are the original periodic hold and step, 60 ms sign dwell (120 ms period), irregular alternation, seeded random targets, and biased targets. The four new streams have zero goal, a 3.5 s deadline, and a 4 s horizon. Requests, feedback, limits, three rates, and two disturbances are shared. Inputs and implementation were frozen before physical runs. Every FIR guard accepts; removing it leaves the same reference.

Independent 80-digit replay checked 144 cases at both output resolutions: 1,502,784 trace rows and 21,288 event rows. All 144 endpoint tests pass; only 66 full-task tests pass. All 108 smooth-method cases pass the sampled position band $|x - q| \le .2$ mm, not a test of the full transformed cube. These are numerical results for the declared polynomial references, not additional Lean proofs.

Peak physical hold error (mm)	Raw	Fast Ruckig	Min. .8 s	FIR
Periodic, 40 ms	.123–.136	.986–1.014	.034–.048	.033–.046
60 ms sign dwell	.515–.544	1.000–1.020	.075–.090	.103–.115
Irregular	.796–.817	.960–.997	.178–.192	.423–.435
Seeded random	.432–.445	.891–.923	.089–.102	.234–.248
Biased	.411–.427	.986–1.017	.329–.342	.400–.413

Ranges use the .5 ms traces and event samples across six plant settings. The hold limit is .2 mm. The irregular stream has zero arithmetic mean but a time-weighted mean of $-.384615$ mm; unequal dwell times create bias. The random stream is one fixed realization, not a distribution-wide result. Every method fails the deliberately biased stream.

A concrete filter failure. With random targets, $r = 80\ \mathrm{s}^{-1}$, $w = .01\ \mathrm{m/s}^2$ and $t = .896$ s, $x = .247558$ mm and $q = .234756$ mm. Tracking error is only .012802 mm, yet the hold band fails. The minimum-duration baseline passes. For the 20 mm step, settling after commitment is .7975–.7980 s for that baseline and .7435–.7440 s for FIR. FIR uses less jerk energy. No method wins every measure.

The original execution check failed. Two of 12 native checks have acceleration errors of 3.57×10^{-9} and $7.42 \times 10^{-8}\ \mathrm{m/s}^2$, above the unchanged 10^{-10} limit. The second occurs on an ordinary grid point after repeated replans. The canonical path carries polynomial state at rational times; native execution carries its own state after double clock subtraction.

A separate repair. Gate 06 uses native retained state and binary execution origins, with unchanged inputs and limits. Independent preflight matches all 492 plans and 12 exported schedules. Independent 80-digit replay of 1,514,064 trace rows and 26,928 event rows reproduces unchanged task decisions. The finite native checks pass the original tolerances; they do not prove firmware fidelity. The old failed gate remains preserved.

A metric can change while motion barely changes. Total variation counts absolute changes, including jumps. For $\sigma = 10\ \mathrm{s}^{-1}$ and SI units, a change at fixed physical state gives

$$\Delta u = 800\Delta q + 60\Delta \dot{q} + 1.5\Delta \ddot{q} + \Delta j/96.$$

Two opposite 15 $\mathrm{m/s}^3$ jerk jumps add 5/16 $\mathrm{m/s}^2$ of command variation, even around a sub-picosecond phase. In biased fast Ruckig, ten such phases become five: jump variation changes by $-25/16\ \mathrm{m/s}^2$, with a state-join correction near $-1.65 \times 10^{-16}\ \mathrm{m/s}^2$. Continuous variation is separate. This exact audit does not establish a comparable motion change or sampled-actuator behavior.

7 Tracking through rounded reference handoffs

The same physical argument can cover an exported native path. It needs a different reference contract. The export consists of cubic position segments with constant jerk. Floating-point coefficients can leave small position, velocity, and acceleration jumps at joins. We retain those jumps; we do not replace them with exact matches.

Let $Q_k = (q_k, \dot{q}_k, \ddot{q}_k)$ be segment k, with local duration d_k. Define the linear map

$$T_\sigma(y_0, y_1, y_2) = \left(y_0,\ y_0 + \frac{y_1}{4\sigma},\ y_0 + \frac{y_1}{2\sigma} + \frac{y_2}{16\sigma^2}\right).$$

For each join, bound

$$\left\|T_\sigma\big(Q_{k+1}(0) - Q_k(d_k)\big)\right\|_\infty \le B_k, \qquad B_k \ge 0.$$

Theorem 3 (Finite handoffs with a reference jump reserve)**.** *Let $\sigma > 0$. Consider finitely many cubic reference segments with $d_k \ge 0$. On every segment, assume the physical dynamics and feedback of (2)–(3), the rate and disturbance bounds, and the local segment jerk bound $|j_k| \le J$. The physical state is continuous on each segment and matches exactly at joins. Derivatives are required only in segment interiors. Assume the initial transformed error is at most $R_0 \ge 0$ and*

$$W + J/(48\sigma) \le 2\sigma^2 R_0.$$

Then, throughout segment k, including its endpoints,

$$\|T_\sigma(x - q_k, v - \dot{q}_k, a - \ddot{q}_k)\|_\infty \le R_0 + \sum_{i<k} B_i.$$

At a join, the transformed tracking error changes by minus the transformed reference jump. The triangle inequality adds B_k to the radius. The larger radius still satisfies the segment budget. Lean derives each segment's propagation from its physical trajectory equations, then proves the finite induction. No segment tracking bound is assumed as a new premise.

Binding the native comparison. A Lean checker verifies exact rational encodings of the frozen comparison's 12 Ruckig reference schedules: 886 cubic pieces and 874 joins. It checks positive duration, adjacent clock equality, initial rest, final horizon, $|j_k| \le 7.5$, and all three transformed jump components. The coefficients are the exact binary values stored in the export. Clock endpoints use the recorded exact values. An independent value audit binds every encoded piece to the frozen export. The JSON-to-Lean parser is not kernel-verified. Lean proves that an accepted check supplies the real cubic theorem's data premises. It also proves that the intervals cover every real time in the horizon.

With the unchanged $\sigma = 10\ \mathrm{s}^{-1}$, $W = .01\ \mathrm{m/s}^2$, and $J = 7.5\ \mathrm{m/s}^3$, use

$$R_0 = \frac{41}{320000}\ \mathrm{m}, \qquad B_k = 10^{-15}\ \mathrm{m}.$$

There are at most 189 pieces in any schedule. The common position tracking bound is therefore

$$|x - q| \le R_0 + 188 \cdot 10^{-15}\ \mathrm{m} = 0.128125000188\ \mathrm{mm} < 0.2\ \mathrm{mm}.$$

The reserve was selected after inspecting the frozen exports. It is a conservative proof allowance, not a new performance result. The exact rational audit found a largest accumulated transformed jump of about $7.39 \cdot 10^{-18}$ m.

Scope. The 12 named Lean instances concern these exact exported paths under the stated physical assumptions. The reference need not have a global classical third derivative at joins. These results do not certify the native solver, floating-point execution, every possible Ruckig output, or an industrial arm. They prove tracking relative to the reference. All earlier true-goal failures remain. A task guarantee still needs a bound on $q - g$; physical rest and tight endpoint speed or acceleration do not follow from this position bound.

8 Human motion and industrial execution

The practical question is how to preserve smooth execution while a planner changes its request. Smooth robot motion already exists. A useful comparison must include an industrial controller's native interpolation and blending, with its timing and accuracy costs.

Industrial reference points. The public description of FANUC's termination guide compares FINE and continuous motion. It discusses how blending, speed, approach angle, and payload affect path accuracy and cycle time [9]. Its official ROS 2 driver guide warns that PC timing jitter can cause drift or vibration in the documented 1 kHz configuration; buffer depth and update rate affect that interface [10]. The guide's login-gated video was not viewed. These are configuration-specific facts, not a public model of the proprietary servo law.

Yaskawa distinguishes buffered targets, point-to-point requests that require full deceleration, and external increments supplied at the interpolation clock. Buffered YMConnect uses the INFORM planner, and the guide warns that very dense points can disturb the velocity profile [11]. The SMOOTH function changes motion filtering. Its documented examples show effects on torque and acceleration, later attainment of speed, and more inward corner rounding [12]. We have not measured these industrial systems or reconstructed their internal algorithms.

A fair hardware comparison would use the same arm, load, task, path tolerance, deadline, and declared motion limits. It would explicitly identify and vary the approved execution interface: native blended motion, buffered external targets, and the proposed execution rule on its supported interface. Record command age, buffer occupancy, goal changes, actual path error, vibration, and settling. A sequence of forced intermediate stops is not an adequate substitute for the native blended baseline.

Human motion as a source of hypotheses. Minimum-jerk models reproduce features of human reaching in specific tasks [13]. Optimal feedback control gives another useful principle: correct deviations that threaten the task while permitting harmless variation [14]. Neither establishes that a brain runs this filter or that robot jitter has a neurological cause. The engineering hypothesis is narrower: repeated target estimates may impose unnecessary motion obligations when the intended task has not changed.

Smoothness must survive a new plan. A low jerk cost within each plan does not bound the joins between plans. Penalizing snap also needs a stated treatment of jerk discontinuities. Repeated planning can be useful, but the executor must account for its present state, committed motion, command age, and the meaning of a revision. A changed estimate and a changed goal are different inputs.

Energy is a separate metric. Signed mechanical work, positive work, electrical energy, squared torque, and squared jerk measure different things. For an ideal frictionless point mass,

$$\int_0^T F(t)v(t)\,dt = \tfrac{1}{2}m\,[v(T)^2 - v(0)^2].$$

Every rest-to-rest path has zero net work in this model, including rough ones. Thus net work alone cannot select smooth motion. Our scalar comparison has no motor loss or electrical supply model. Command variation and jerk cost must not be reported as measured energy savings.

Next controlled comparison. Separate authoritative goal revisions from advisory estimates, and give every baseline the same information. The frozen plan keeps the four existing scalar methods and inputs. A later, separately specified comparison should test limits on jerk, jerk or snap costs, and continuity across plan replacement at the same accuracy and deadline. Industrial hardware comparisons require their own interface and measurement protocol. An always-hold rule must fail an intended move; a smoother but late move must retain its failure. The task-contract study specifies a bounded position-progress candidate, but it is not yet a proved full endpoint guarantee or an executed new comparison.

9 Existing work and the remaining research claim

FIR trajectory generation and derivative constraints are established. Biagiotti and Melchiorri study FIR planning and time optimization [2, 4]. Besset and Béarée study online jerk-constrained FIR trajectories, including nonzero initial states and robot experiments [3]. Our four-box kernel is not a new smoothing method.

Real-Time Chunking retains committed actions while generating the remaining chunk [5]. VLA-RAIL uses polynomial smoothing and chunk fusion with position, velocity, and acceleration continuity [6]. These close works prevent a broad claim to introduce a VLA smoothing layer. Our scalar experiment does not compare task performance against them.

Reference and command governors already use finite checks for future constraints, invariant fallbacks, and uncertainty bounds. The survey also treats asynchronous execution and plant-side acceptance [7]. ModelPlex derives checked arithmetic monitors from proved models and addresses bounded disturbances, clock drift, and arithmetic implementation [8]. These works rule out a broad claim to introduce future checking or a verified execution layer.

The narrower candidate is this list-parametric Lean rational guard, its retained-history contract, its connection to a stated uncertain tracking bound, and a checked bridge for the frozen rounded native references. This is a specialized formalization and engineering case study. It is not yet an established new control method. The comparison ledger records which full methods were read and which FIR sources were only partly accessible. External specialist review is still needed to judge the artifact's research value.

Limits on task intent. If two tasks give the same observed history, a deterministic causal executor gives the same motion until the histories differ. If the tasks require goals g_1 and g_2, then

$$|g_1 - g_2| \le |x - g_1| + |x - g_2|.$$

At least one error is at least half the goal separation. This elementary argument is not a new theorem. It explains why a noise policy needs an intent or delay contract.

Next publication gate. Make the proof artifact easy to replay, preserve the negative cases, and seek human formal-methods and controls review before expanding the claim. A real action-stream test must report delay, task error, actual motion, and failures together. A scalar position request is not a full robot action chunk. Joint coupling, contact, sampled actuation, and task progress need further work.

Reproducibility. See `research/pods/2026-10-17/robot-motion/`: `certificate-03/`, `comparison-04/`, `latency-noise-05/`, `native-execution-06/`, `rounded-handoff-07/`, `intent-contract-07/`, and `jerk-prefix-animation/`. The FIR proof bundle is in `publication/robot-motion/`, under `reviewer-bundle/`. Lean is version 4.35.0. The mathlib and Ruckig pins are:

```
56a005855ea17e0b0958f73ec801ae140b0a1679
99751e6ebc7c963bb80e99fbbe66f2c279d1db57
```

Ruckig cloud support is disabled.

Numerical replay treats exported floating coefficients as exact polynomial inputs and retains tiny positive phases. It is not bit-for-bit native execution. Switch-side differences remain in the reports. Physical states carry continuously from rest. Maxima and quadrature are numerical metrics.

Contributions and assistance. Mike Helsel set the question, proposed the execution-layer direction, identified the timing interpretation to test, required fair comparisons and independent challenge, and proposed the links to task intent, human motion, industrial interpolation, and smoothness or effort costs. Codex agents assisted with sources, derivations, Lean proofs, numerical code, reviews, figures, and writing. Agent reviews are computational checks, not external human peer review. Author details, license, venue policy, and final claims remain to be settled before submission.

References

[1] L. Berscheid and T. Kröger. *Jerk-limited Real-time Trajectory Generation with Arbitrary Target States*. RSS, 2021. https://arxiv.org/abs/2105.04830v2.

[2] L. Biagiotti and C. Melchiorri. *FIR Filters for Online Trajectory Planning with Time- and Frequency-Domain Specifications*. Control Engineering Practice 20(12), 1385–1399, 2012. https://doi.org/10.1016/j.conengprac.2012.08.005.

[3] P. Besset and R. Béarée. *FIR filter-based online jerk-constrained trajectory generation*. Control Engineering Practice 66, 169–180, 2017. https://doi.org/10.1016/j.conengprac.2017.06.015.

[4] L. Biagiotti and C. Melchiorri. *Trajectory generation via FIR filters: A procedure for time-optimization under kinematic and frequency constraints*. 2019. https://iris.unimore.it/handle/11380/1175126.

[5] K. Black, M. Y. Galliker, and S. Levine. *Real-Time Execution of Action Chunking Flow Policies*. NeurIPS, 2025. https://arxiv.org/abs/2506.07339v2.

[6] Y. Zhao et al. *VLA-RAIL: A Real-Time Asynchronous Inference Linker for VLA Models and Robots*. 2025. https://arxiv.org/abs/2512.24673v1.

[7] E. Garone, S. Di Cairano, and I. Kolmanovsky. *Reference and command governors for systems with constraints: A survey on theory and applications*. Automatica 75, 306–328, 2017. https://doi.org/10.1016/j.automatica.2016.08.013.

[8] S. Mitsch and A. Platzer. *ModelPlex: Verified Runtime Validation of Verified Cyber-Physical System Models*. Runtime Verification, 199–214, 2014. https://symbolaris.com/pub/modelplex.pdf.

[9] FANUC America. *How Termination Type Affects Robot Path and Cycle Time*. Public guide description, 18 June 2026; video not viewed. https://techtransfer.fanucamerica.com/tech-transfer/termination-types.

[10] FANUC Corporation. *How to Configure 1kHz Mode*. ROS 2 Driver Documentation, v2.2.0. https://fanuc-corporation.github.io/fanuc_driver_doc/v2.2.0/docs/fanuc_driver/how_to_configure_1khz_mode.html.

[11] Yaskawa Motoman. *Explanation of Different Remote-Motion-Control API's*. Updated 14 November 2024. https://knowledge.motoman.com/hc/en-us/articles/20684798540951-Explanation-of-Different-Remote-Motion-Control-API-s.

[12] Yaskawa Motoman. *SMOOTH Function Overview*. Updated 28 July 2023. https://knowledge.motoman.com/hc/en-us/articles/16313283895191-SMOOTH-Function-Overview.

[13] T. Flash and N. Hogan. *The coordination of arm movements: an experimentally confirmed mathematical model*. Journal of Neuroscience 5(7), 1688–1703, 1985. https://doi.org/10.1523/JNEUROSCI.05-07-01688.1985.

[14] E. Todorov and M. I. Jordan. *Optimal feedback control as a theory of motor coordination*. Nature Neuroscience 5, 1226–1235, 2002. https://doi.org/10.1038/nn963.